SEVACARE AI

GOVERNMENT HEALTHCARE · AI-POWERED

SevaCare AI

Transforming Government Hospital Queues with Artificial Intelligence

SevaCare AI is an intelligent, end-to-end appointment and queue management platform purpose-built for government hospitals across India. By combining AI-driven queue prediction, smart scheduling, and real-time patient notifications, SevaCare AI eliminates the chaos of overcrowded waiting rooms — delivering a faster, fairer, and more dignified healthcare experience for millions of citizens every single day.

THE CHALLENGE

The Problem: A Healthcare System at Capacity

Government hospitals across India serve millions of patients daily, yet the infrastructure for managing patient flow has remained largely unchanged for decades. Manual token systems, overcrowded waiting halls, and zero real-time communication between patients and hospital staff create a cascade of inefficiencies that harm both patients and healthcare workers. The result is a system where a routine check-up can consume half a day, where doctors are overwhelmed by disorganized patient loads, and where critical cases go unnoticed in the crowd. This is not just an inconvenience — it is a systemic crisis affecting healthcare delivery at scale.

Patients arrive early, wait for hours with no visibility into their position in the queue, and leave frustrated — many without ever being seen. Doctors face unpredictable patient volumes, making it nearly impossible to plan their schedules or allocate adequate time per case. Emergency cases get buried in general queues. Hospital administrators have no real-time data to make informed staffing or resource decisions. The absence of digital infrastructure means that even basic queue management relies on paper tokens and verbal announcements — a model that simply cannot scale to meet the demands of a growing population.

⏱️ Long Wait Times

Patients wait 3–5 hours on average for a routine consultation, with no visibility into queue progress or estimated wait duration.

📋 Manual Token Systems

Paper-based token distribution creates confusion, errors, and zero traceability — impossible to scale or audit.

🩺 Doctor Overload

Physicians face unpredictable patient volumes with no intelligent load balancing, leading to burnout and reduced quality of care.

📢 Zero Communication

Patients receive no updates, no reminders, and no guidance — creating anxiety, no-shows, and overcrowding in waiting areas.

3.8h

Avg. Wait Time

Per patient visit at government hospitals without digital queue management

72%

Patient Dissatisfaction

Report frustration with current appointment and queue systems

40%

Doctor Burnout Rate

Among physicians in high-volume government hospital settings

1 in 3

Patients Leave

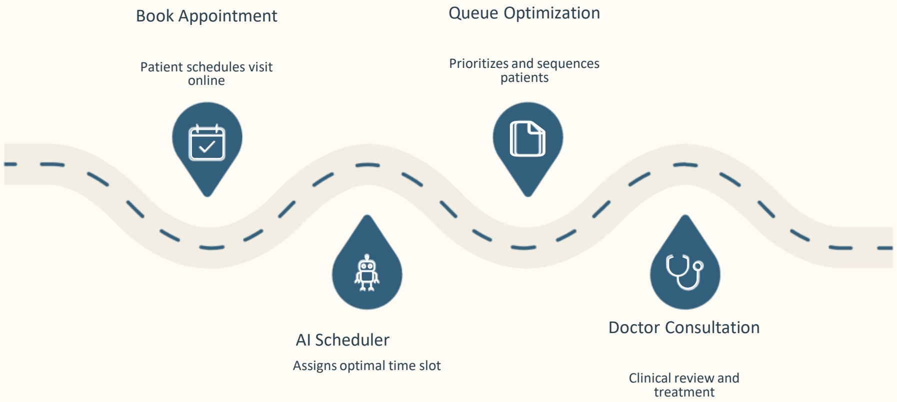
Without being seen due to excessive waiting times

OUR SOLUTION

How SevaCare AI Solves This

SevaCare AI is a comprehensive, AI-powered platform that reimagines the entire patient journey — from the moment a patient decides to visit a hospital, through check-in, queue management, consultation, and follow-up. The system replaces manual, paper-based processes with intelligent digital workflows that are transparent, efficient, and patient-centric. By integrating smart appointment booking, real-time queue prediction, live notifications, and doctor load optimization into a single unified platform, SevaCare AI ensures that every stakeholder — patients, doctors, and administrators — has the information and tools they need to make the healthcare experience smoother and faster.

At its core, SevaCare AI uses machine learning models trained on historical patient flow data to predict queue wait times with high accuracy, dynamically balance patient loads across available doctors, and proactively communicate updates to patients via SMS and in-app notifications. The platform is designed to integrate seamlessly with existing hospital infrastructure, requiring minimal hardware changes while delivering maximum operational impact. Whether a patient is booking an appointment from home or walking in without one, SevaCare AI ensures they are guided, informed, and served efficiently.



This end-to-end workflow ensures that no patient is left waiting without information, no doctor is overwhelmed without warning, and no administrator is making decisions without data. SevaCare AI transforms a chaotic, manual process into a streamlined, intelligent system — reducing wait times, improving patient satisfaction, and enabling healthcare workers to focus on what matters most: delivering quality care.

For Patients

- Book appointments online or via walk-in
- Receive real-time queue position updates
- Get estimated wait time notifications
- Emergency cases automatically prioritized

For Hospitals

- AI-powered queue prediction and load balancing
- Live admin dashboard with full visibility
- Doctor schedule optimization
- Data-driven resource planning

PLATFORM FEATURES

Key Features of SevaCare AI

SevaCare AI is built with a comprehensive suite of features that address every pain point in the government hospital patient journey. From intelligent appointment booking to real-time queue dashboards, AI-powered notifications, and an always-available helpdesk chatbot — every feature is designed to reduce friction, increase transparency, and improve outcomes for patients and healthcare providers alike. The platform is modular, scalable, and deployable across hospitals of varying sizes and patient volumes.

Online Appointment Booking

Patients can book, reschedule, or cancel appointments online — reducing walk-in congestion and enabling better doctor schedule planning across departments.

AI Queue Prediction

Machine learning models analyze historical and real-time patient flow data to accurately predict wait times and optimize queue distribution dynamically.

Live Queue Dashboard

Real-time visibility into current token numbers, department-wise queue density, and doctor availability — accessible to patients and administrators simultaneously.

Smart Scheduling

Intelligent slot allocation based on doctor availability, patient urgency, and predicted queue load — ensuring optimal utilization of clinical resources.

Notification Alerts

Automated SMS and in-app notifications keep patients informed about their queue position, estimated wait time, and when to arrive at the hospital.

AI Helpdesk Chatbot

24/7 conversational AI assistant that answers patient queries, guides appointment booking, and provides hospital information in multiple regional languages.

Emergency Prioritization

Critical cases are automatically identified and elevated in the queue — ensuring urgent patients receive immediate attention without disrupting the general flow.

Admin Dashboard

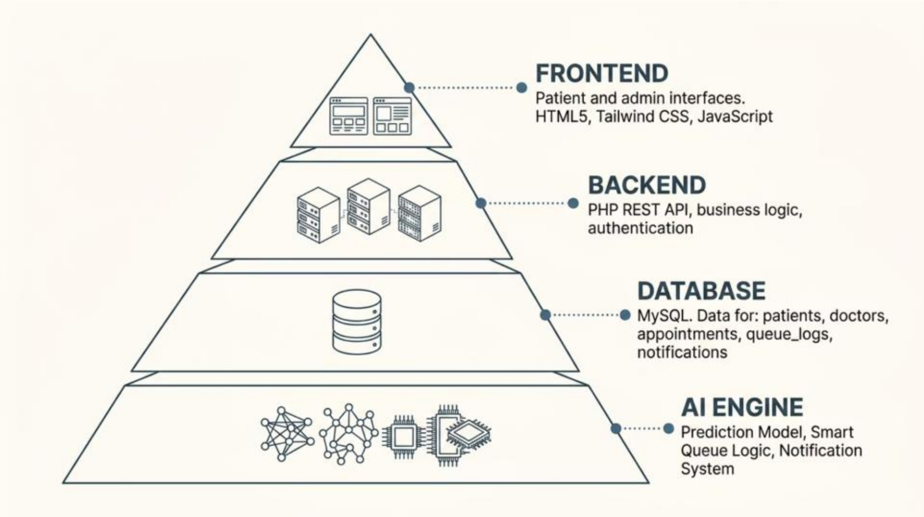
A centralized control panel for hospital administrators to monitor operations, manage doctor rosters, analyze trends, and generate compliance reports.

TECHNICAL ARCHITECTURE

System Architecture & Technology Stack

SevaCare AI is built on a modern, layered architecture designed for reliability, scalability, and ease of deployment in government hospital environments. The platform separates concerns cleanly across frontend, backend, database, and AI layers — ensuring that each component can be updated, scaled, or replaced independently without disrupting the overall system. The technology choices prioritize open-source tools, low infrastructure costs, and compatibility with existing hospital IT environments, making SevaCare AI practical for large-scale public sector deployment.

The frontend is built with HTML5, Tailwind CSS, and vanilla JavaScript — delivering a responsive, mobile-first user interface that works seamlessly across devices from low-end smartphones to desktop workstations. The backend leverages PHP for server-side logic, chosen for its widespread hosting support and compatibility with government server environments. MySQL serves as the primary database, providing robust relational data management for patient records, appointments, queue logs, and notifications. The AI layer operates as an independent prediction engine, processing queue data through trained models and returning optimized scheduling recommendations in real time.



The AI prediction engine is the intelligence core of SevaCare AI. It continuously ingests queue data, appointment patterns, doctor availability, and historical patient flow to generate accurate wait-time estimates and dynamic scheduling recommendations. The smart queue logic automatically redistributes patient load when bottlenecks are detected, while the notification system ensures patients are proactively informed of any changes to their appointment status or queue position.

Frontend Layer

- HTML5 — Semantic, accessible markup
- Tailwind CSS — Utility-first responsive design
- JavaScript — Dynamic UI and API interactions

Backend & AI Layer

- PHP — Server-side logic and REST API
- MySQL — Relational database for all entities
- AI Engine — Prediction, queue logic, notifications

PRODUCT SHOWCASE

Website UI & Live Queue Dashboard

SevaCare AI features a clean, intuitive user interface inspired by modern SaaS dashboards — designed to be accessible to patients of all technical backgrounds while providing powerful tools for hospital administrators. The UI follows a consistent design language with rounded cards, subtle shadows, and a healthcare blue palette that conveys trust and professionalism. Every screen is optimized for both desktop and mobile, ensuring that patients can interact with the system whether they are at home or in the hospital waiting area.

The homepage provides a clear entry point for patients to check live queue status, book appointments, or access the AI helpdesk — all within two taps. The live queue dashboard displays real-time token numbers, estimated wait times, and department-wise traffic in a visually intuitive format. The admin dashboard gives hospital staff complete operational visibility, including doctor schedules, queue analytics, and patient flow trends. The booking system guides patients through a simple, step-by-step appointment flow with smart slot suggestions based on AI predictions.

Homepage & Booking

A clean, patient-friendly entry point with instant access to appointment booking, live queue status, and the AI helpdesk — all visible within seconds of landing on the site.

Live Queue Dashboard

Real-time token tracking, estimated wait times, and department-wise queue density — giving patients and administrators complete visibility into current hospital flow.

Admin Dashboard

A centralized command center for hospital administrators — manage doctor rosters, monitor queue performance, analyze patient trends, and generate operational reports.

AI INTELLIGENCE

Live Queue Prediction & AI Features

At the heart of SevaCare AI is a sophisticated prediction engine that transforms raw queue data into actionable intelligence. By analyzing historical patient flow patterns, current appointment volumes, doctor availability, and department-specific trends, the AI generates accurate, real-time wait-time estimates that are continuously updated as conditions change. This is not a static system — it learns and adapts, becoming more accurate over time as it processes more data from daily hospital operations.

The AI layer operates across multiple dimensions simultaneously: it balances patient loads across available doctors to prevent bottlenecks, identifies emergency cases that require immediate prioritization, predicts peak traffic windows to enable proactive staffing adjustments, and communicates all relevant updates to patients through automated notifications. The result is a system that feels intelligent and responsive — anticipating problems before they occur and guiding both patients and staff toward the most efficient outcomes.

Token #47

Current Token

Active patient being served in General Medicine department right now

22min

Est. Wait Time

AI-predicted wait for next patient based on live queue data

8

Active Doctors

Currently available across all departments in the hospital

142

Patients Today

Total appointments and walk-ins processed so far today

Smart Queue Balancing

Dynamically redistributes patient load across doctors to minimize bottlenecks and ensure even utilization of clinical resources throughout the day.

Doctor Load Optimization

Monitors physician workload in real time and suggests schedule adjustments to prevent burnout while maintaining patient throughput efficiency.

Emergency Prioritization

Automatically identifies and elevates critical cases in the queue, ensuring urgent patients bypass standard wait times without disrupting overall flow.

AI Chatbot Assistant

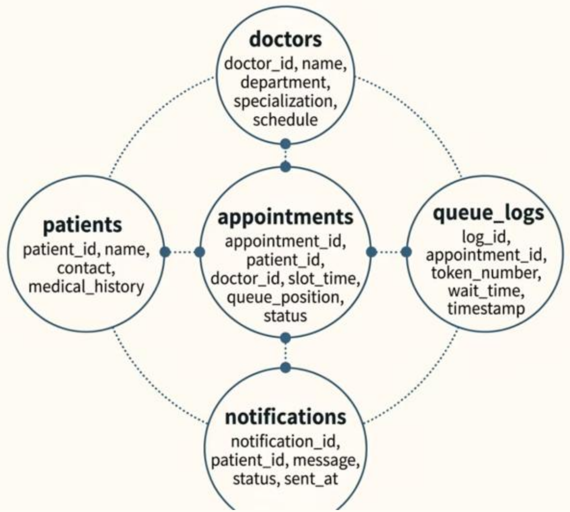
A conversational AI helpdesk available 24/7 to answer patient queries, guide appointment booking, and provide hospital information in regional languages.

DATA ARCHITECTURE

Database Design & Data Model

SevaCare AI's data architecture is built around five core entities that together capture the complete patient journey — from registration and appointment booking through queue management, consultation, and notification delivery. The database schema is designed for high query performance under heavy concurrent load, with normalized tables that eliminate redundancy while maintaining referential integrity across all relationships. Each entity is optimized for the specific read and write patterns it experiences during daily hospital operations.

The **patients** table stores demographic information, contact details, and medical history references, serving as the primary identity record for every individual in the system. The **doctors** table maintains physician profiles, department assignments, specialization data, and availability schedules. The **appointments** table is the central transactional entity, linking patients to doctors with scheduled time slots, appointment status, and queue position data. The **queue_logs** table provides a complete audit trail of every queue event — token generation, position changes, wait times, and service completion — enabling both real-time operations and historical analytics. Finally, the **notifications** table tracks every message sent to patients, including delivery status and interaction data, ensuring full accountability for patient communications.



This relational model ensures that every piece of data in SevaCare AI is connected, traceable, and queryable. The appointments table acts as the central hub, linking patient records to doctor schedules while feeding queue logs and triggering notification events. This architecture enables complex analytical queries — such as calculating average wait times by department, identifying peak traffic patterns, or measuring doctor utilization rates — without requiring separate data warehouses or ETL processes. The schema is also designed to scale horizontally, supporting deployment across multi-hospital networks as the platform expands.

Core Entities

- patients — Identity and medical records

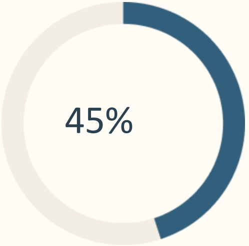
Operational Tables

- queue_logs — Complete queue event audit trail

IMPACT & ROADMAP

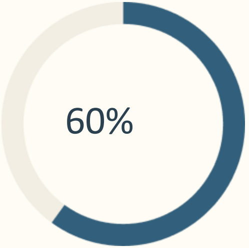
Real-World Impact & Future Scope

SevaCare AI is designed to deliver measurable, quantifiable improvements in hospital operational efficiency and patient experience. Based on projected deployment models and comparable AI-driven queue management systems implemented in public healthcare settings, the platform is expected to deliver significant reductions in patient wait times, improvements in queue throughput, and meaningful decreases in overcrowding. These projections are grounded in real-world benchmarks from similar digital transformation initiatives in government hospitals, and they represent conservative estimates that improve as the AI model learns from live operational data over time.



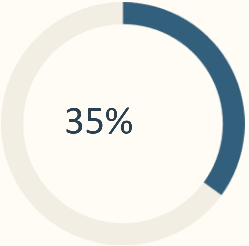
Reduced Wait Time

Projected decrease in average patient waiting time through AI-powered queue optimization and smart scheduling



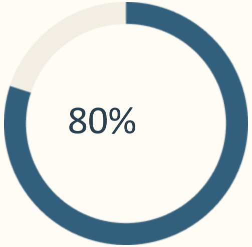
Queue Efficiency Gain

Expected improvement in overall queue throughput and patient handling capacity across departments



Less Overcrowding

Anticipated reduction in waiting hall congestion through distributed appointment slots and live notifications



Patient Satisfaction

Target satisfaction score based on improved communication, transparency, and reduced uncertainty

Beyond the initial deployment, SevaCare AI has a clear and ambitious roadmap for expansion. The platform is architected to support additional modules that extend its capabilities from queue management into a comprehensive digital healthcare ecosystem. Each future module is designed to integrate natively with the existing data model, ensuring a seamless upgrade path for hospitals already using the core platform. The long-term vision is a statewide, and eventually national, network of connected government hospitals sharing AI-driven insights and operational intelligence.

Phase 1 — Launch 1

THANK YOU

Building Smarter Healthcare Infrastructure with AI

> SevaCare AI is more than a final year project — it is a vision for what government healthcare can become when technology is applied with purpose, empathy, and intelligence. We are building the infrastructure that millions of citizens deserve.

We believe that every patient who walks into a government hospital deserves the same level of care, respect, and efficiency as those who visit private facilities. SevaCare AI is our commitment to that belief — a platform that uses the power of artificial intelligence to bridge the gap between resource-constrained public hospitals and the growing expectations of the communities they serve. This is not just a software project; it is a step toward a more equitable, efficient, and humane healthcare system for all.

We welcome questions, feedback, and collaboration opportunities from investors, hospital administrators, and academic reviewers who share our vision for digital transformation in public healthcare. SevaCare AI is ready for pilot deployment, and we are eager to demonstrate its impact in real hospital environments. Together, we can build the smarter healthcare infrastructure that India needs and its citizens deserve.

📧 Contact Us

info@sevacare.ai
hello@sevacare.ai

🌐 Website

www.sevacare.ai
Live Demo Available

📱 Connect

GitHub · LinkedIn
Research Paper

🏛 Project

Final Year Engineering Project
Department of Computer Science

Project Contributors

Final Year Engineering Team · Computer Science Department · Academic Year 2024–25

Powered By

AI · Machine Learning · PHP · MySQL · Tailwind CSS · Modern Healthcare Tech

Made With Purpose

Designed for Government Hospitals · Built for Scale · Deployed for Impact